CPP NOTES – DAY 26

ABSTRACTION

Abstraction is the process of hiding complex implementation details and showing only the essential features of an object to the user. In simple terms, you focus on *what* an object does, not *how* it does it.

ENCAPSULATION

Encapsulation is another core principle of Object-Oriented Programming (OOP). It refers to the bundling of data (variables) and functions (methods) that operate on the data into a single unit (a class), and restricting direct access to some of the object's components.

POLYMORPHISM

Polymorphism is a fundamental concept in Object-Oriented Programming (OOP). The term means "many forms." It allows the same function name, operator, or method call to behave differently depending on the context—typically based on the object or arguments involved.

INHERITANCE

Inheritance is one of the key pillars of Object-Oriented Programming (OOP). It allows one class (called a derived class) to inherit properties and behaviors (members) from another class (called a base class).

FYI:

- *When a structure is declared it is saved in the text block of memory.*
- *Que: If all the variable and methods of a class are private what will happen to the class/importance of the class? Ans: Then the class becomes inaccessible from the outside.*
- *string class ---> array of characters -> c_str() is used.*
- *One method calling another method inside the class is said to be nested methods*

- *:: → is the scope resolution operator(can be used when declaring function prototypes inside the class and defining the function outside with this scope.*
- *Class should be defined in the header file (inc) and that can be included in the cpp file.*